Three colours and three generations from perturbativity in 3-3-1 models

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Abstract

In $SU(3)_C \times SU(3)_L \times U(1)_X$ ("3-3-1") gauge models, anomaly cancellation ties the number of fermion generations to the number of colours. We keep the number of colours N_c general throughout, with every charge and anomaly coefficient computed as a function of N_c, and add a single requirement: no gauge coupling may become non-perturbative below the Planck scale. The $SU(3)_L$ anomaly together with asymptotic freedom of colour forces N_c to be odd and $N_g = N_c$, which makes Witten's global anomaly condition automatic. $SU(3)_L$ running then excludes $N_c \geq 7$ and $U(1)_X$ running excludes $N_c = 5$, so that $N_c = N_g = 3$ is the only survivor, with quark charges $\frac{2}{3}$ and $-\frac{1}{3}$. The selection holds at two loops in the gauge couplings with two-loop Yukawa terms, with one-loop threshold matching across the heavy spectrum, for 3-3-1 breaking scales from 2 to 30 TeV, and in both variants $\beta = \pm 1/\sqrt{3}$. It takes the measured electroweak couplings at the TeV scale as input and does not predict them; in the $\beta = +1/\sqrt{3}$ variant the $N_c = 3$ margin is a factor of about 2 in the pole scale. Requiring the scalar sector itself to remain perturbative and bounded below up to $M_{\rm Pl}$ then bounds the exotic-quark masses relative to the bilepton mass. All numbers are reproduced by open code with regression tests.

1 Introduction

In 3-3-1 models [1, 2] the $SU(3)_L$ anomaly cancels only between generations, which links the number of generations to the number of colours: N_g must be a multiple of N_c [3, 4, 5]. With the number of light neutrinos taken from experiment, Frampton noted that this singles out $N_c = 3$ [6]. Here no generation or neutrino number is taken from experiment. We ask instead which N_c is compatible with all gauge couplings remaining perturbative up to the Planck scale, given the measured electroweak couplings at the TeV scale. The answer is $N_c = N_g = 3$ only.

Section 2 sets up the model for general N_c. Section 3 gives the four-step selection; Sections 4–6 test its robustness, including one-loop threshold matching and the scalar sector. The conditions under which it holds are collected in Section 7, and Section 8 compares with prior work.

2 Setup for general N_c

The electric charge is $Q = T_3 + \beta T_8 + X$. The two variants without low-scale Landau poles are $\beta = \pm 1/\sqrt{3}$; the minimal $\beta = -\sqrt{3}$ model is excluded by atomic parity violation and sits at the LHC Z' bound (Section 9). For colour group $SU(N_c)$ we take N_g generations. Each has left-handed lepton triplets and quark triplets or anti-triplets, the third member of each quark multiplet being an exotic quark; n of the N_g quark generations sit in the anti-triplet.

Cancelling the perturbative anomalies, with the Yukawa couplings in place, gives

$$Q_u = \frac{N_c + 1}{2N_c}, \qquad Q_d = \frac{1 - N_c}{2N_c}, \tag{1}$$

which reduce to $\frac{2}{3}$ and $-\frac{1}{3}$ at $N_c = 3$. Every gauge anomaly (X^3, grav^2X, $SU(3)_L^2X$, $SU(N_c)^2X$, $SU(3)_L^3$) then vanishes identically in N_c, both in the Standard Model and in the full 3-3-1 content; this was checked in exact rational arithmetic for $N_c = 3, 5, 7$. Perturbative anomalies therefore fix the charges given N_c, but not N_c itself.

The $U(1)_X$ charges for $\beta = -1/\sqrt{3}$ are $X = Q_u - \frac{1}{3}$ (quark triplet), $Q_d + \frac{1}{3}$ (anti-triplet), $-\frac{1}{3}$ (leptons), and $-\frac{1}{3}, +\frac{2}{3}, -\frac{1}{3}$ for the scalars η, ρ, χ, fixed by the Yukawas and independent of N_c. For $\beta = +1/\sqrt{3}$: leptons $-\frac{2}{3}$, quark triplet $Q_u - \frac{2}{3}$, anti-triplet $Q_d + \frac{2}{3}$, scalars $\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}$.

We require a non-abelian colour group, $N_c \geq 2$: for $N_c = 1$ there is no colour gauge field at all, and the asymptotic-freedom condition used below is vacuous.

3 The selection

Step 1: $N_g = N_c$. $SU(3)_L$ anomaly freedom requires equal numbers of triplets and anti-triplets,

$$N_g(1 + N_c) = 2nN_c. \tag{2}$$

For odd N_c, $\gcd(N_c, N_c + 1) = 1$, so $N_c \mid N_g$ and $N_g = kN_c$. With three quark flavours per generation, asymptotic freedom of $SU(N_c)$ requires $11N_c/3 - 2N_g > 0$, i.e. $k < 11/6$, so $k = 1$. For even N_c, $\gcd(N_c + 1, 2N_c) = 1$, so $2N_c \mid N_g$; but $2N_c > 11N_c/6$ violates asymptotic freedom. Hence N_c is odd and $N_g = N_c$, with no further input. At $N_c = 3$ this reproduces the standard assignment of two quark generations in the anti-triplet and one in the triplet.

Step 2: Witten's anomaly is implied. In the Standard Model, Witten's global $SU(2)$ anomaly [7] requires an even number of doublets, $N_c + 1$ per generation. In 3-3-1, $SU(2)_L \subset SU(3)_L$ and $\pi_4(SU(3)) = 0$, so the UV group imposes no global anomaly of its own. Step 1 already forces N_c odd, and the doublet count $N_g(1 + N_c) = 2nN_c$ is then even automatically. The spin–charge relation $(-1)^F = \exp(i\pi N_c(B - L))$ gives the same condition as a consequence rather than an input (all 29 solutions of (2) up to $N_c = 60$ checked).

Step 3: $SU(3)_L$ running excludes $N_c \geq 7$. With $N_g = N_c$ there are $N_c(1 + N_c)$ Weyl triplets. The one-loop coefficient $b_L = 11 - N_c(1 + N_c)/3 -$ (scalars), run from $1/\alpha_L(4\,\text{TeV}) = 31.5$, gives Table 1. Adding the minimal model's scalar sextet does not change the verdict.

N_c	b_L (3 scalar triplets)	Landau pole
3	$+6.5$	none (asymptotically free)
5	$+0.5$	none
7	-8.2	$1.3 \times 10^{14}\,\text{GeV}$, below M_{Pl}
9	-19.5	$10^8\,\text{GeV}$

Table 1: One-loop $SU(3)_L$ running with $N_g = N_c$.

Step 4: $U(1)_X$ running excludes $N_c = 5$. Two-loop gauge running from the 3-3-1 scale starts at $1/\alpha_X(4\,\text{TeV}) = 83.76$; we define "blow-up" as $g = 4\pi$. The results are in Table 2; the Planck scale is $1.2 \times 10^{19}\,\text{GeV}$. For $\beta = -1/\sqrt{3}$ a pole exactly at M_{Pl} needs $a_X = 14.76$, so the one-loop margins are $+6.09$ for $N_c = 3$ and -4.57 for $N_c = 5$. Two-loop gauge terms move the blow-up scale by a factor of about 3.

$N_c = N_g$	one-loop a_X ($\beta = -1/\sqrt{3}$; $+1/\sqrt{3}$)	blow-up, $\beta = -1/\sqrt{3}$	blow-up, $\beta = +1/\sqrt{3}$
3	$26/3 = 8.67$; 14.00	$\mathbf{3.0 \times 10^{29}}$ GeV	$\mathbf{4.0 \times 10^{19}}$ GeV
5	19.33; 27.33	1.3×10^{15} GeV	5.5×10^{11} GeV
7	34.44; 45.11	9.7×10^{9} GeV	3.1×10^{8} GeV

Table 2: Two-loop $U(1)_X$ blow-up scales.

Two-loop Yukawa terms. A Yukawa coupling $y\,\phi\psi_1\psi_2$ contributes $-g^3y^2\,n_{\text{terms}}\,[C_2(\psi_1) + C_2(\psi_2)]/d(G)/(16\pi^2)^2$ to each gauge β-function [8, 11]; this reproduces the Standard Model top-quark coefficients (17/10, 3/2, 2) exactly. We include the top Yukawa ($y_t = 0.83$ at 4 TeV) and one Yukawa y_J for every exotic quark, and for every heavy lepton when $\beta = +1/\sqrt{3}$, held constant up to the blow-up. Real Yukawas decrease with energy, so this overstates their effect and makes the $N_c = 5$ exclusion conservative. Yukawas raise every pole by at most a factor of about 5 (Table 3), and no constant $y_J \leq 6$ lifts $N_c = 5$ to the Planck scale.

y_J	$\beta = -1/\sqrt{3}$: $N_c = 3$ / $N_c = 5$	$\beta = +1/\sqrt{3}$: $N_c = 3$ / $N_c = 5$
no Yukawas	3.0×10^{29} / 1.3×10^{15} GeV	4.0×10^{19} / 5.5×10^{11} GeV
0 (top only)	3.4×10^{29} / 1.3×10^{15}	4.2×10^{19} / 5.6×10^{11}
1	4.7×10^{29} / 1.6×10^{15}	6.2×10^{19} / 6.8×10^{11}
2	1.2×10^{30} / 2.8×10^{15}	2.2×10^{20} / 1.3×10^{12}

Table 3: Blow-up scales with two-loop Yukawa contributions.

4 Robustness

The two-loop machinery reproduces the Standard Model one-loop coefficients $(41/10, -19/6, -7)$ and all nine two-loop coefficients exactly. Table 4 varies the 3-3-1 scale and the perturbativity criterion.

Check	$\beta = -1/\sqrt{3}$: $N_c = 3$ vs 5	$\beta = +1/\sqrt{3}$: $N_c = 3$ vs 5
3-3-1 scale 2–30 TeV	$(2.7\text{–}3.7) \times 10^{29}$ vs $(0.8\text{–}4.5) \times 10^{15}$	$(2.9\text{–}9.9) \times 10^{19}$ vs $(3.3\text{–}24) \times 10^{11}$
criterion $\alpha = 1$	1.8×10^{29} vs 1.0×10^{15}	2.9×10^{19} vs 4.7×10^{11}
criterion $g = 1$ (strict)	5.3×10^{25} vs 2.7×10^{13}	2.0×10^{17} vs 3.6×10^{10} ($N_c = 3$ fails)

Table 4: Blow-up scales in GeV under variations.

For $\beta = -1/\sqrt{3}$, $N_c = N_g = 3$ is selected under every criterion, with its blow-up 6–10 orders of magnitude above M_{Pl} while $N_c = 5$ fails 4–6 orders below. For $\beta = +1/\sqrt{3}$, 3 is selected under the standard criteria, but its pole lies only a factor 2–3 above M_{Pl} without Yukawas (3–5 with them); under the strict $g = 1$ criterion, $N_c = 3$ fails as well.

5 One-loop threshold matching

The breaking $SU(3)_L \times U(1)_X \to SU(2)_L \times U(1)_Y$ is matched at one loop in $\overline{\text{MS}}$ [12, 13]:

$$\frac{1}{\alpha_{\text{low}}(\mu)} = \frac{1}{\alpha_{\text{high}}(\mu)} - \frac{\lambda}{12\pi}, \quad \lambda = \sum_V t^2\Big(1 - 21\ln\frac{M_V}{\mu}\Big) + 8\sum_{\text{Dirac}} t^2 \ln\frac{M_F}{\mu} + \sum_{\text{real scalars}} t^2 \ln\frac{M_S}{\mu}, \quad (3)$$

with $1/\alpha_X + \beta^2/\alpha_L$ as the high-energy side for $U(1)_Y$. The heavy spectrum is: one complex $SU(2)$ doublet of bilepton vectors ($|Y| = \frac{1}{2}$, mass M_V); the Z' ($M_{Z'} = 1.225\,M_V$); N_c exotic

quarks (plus N_c heavy leptons for $\beta = +1/\sqrt{3}$) of mass $r_J M_V$, whose Yukawa $y_J = r_J g_L/\sqrt{2}$ enters at two loops; and nine heavy real scalars of mass $r_S M_V$ (a second doublet, a charged singlet and three neutral fields), with the option of a light second doublet.

As a check, at one loop the $N_c = 3$ pole must not depend on the matching scale once the spectrum is fixed: moving μ from $M_V/3$ to $3M_V$ changes ln(pole) by less than 10^{-9}, against a factor 2.6 without thresholds. $N_c = 5$ cannot pass this check, because the low-energy theory is the three-colour Standard Model; its pole moves by a factor 1.5 with μ, and we take its most favourable μ.

Scanning $r_J = 0.5$–3, $r_S = \frac{1}{3}$–3, $M_V = 2$–$10\,\text{TeV}$ and a heavy or light second doublet (two-loop running, Yukawas included) gives Table 5. The worst case for $N_c = 3$ (lightest exotics, light scalars, light second doublet, $M_V = 2\,\text{TeV}$) puts the Z' at 2.45 TeV, far below the LHC limits of Section 9. Imposing the most conservative limit from our recast of the full Run-2 dilepton search (Appendix A), $M_{Z'} \geq 4.6\,\text{TeV}$ for $\beta = +1/\sqrt{3}$ and 4.0 TeV for $\beta = -1/\sqrt{3}$, restricts the scan to $M_V \geq 3.76$ and 3.27 TeV respectively. At two loops, moving μ from $M_V/2$ to $2M_V$ changes the pole by 1%.

	$N_c = 3$, worst case	$N_c = 5$, best case	verdict
$\beta = -1/\sqrt{3}$	$2.4 \times 10^{29}\,\text{GeV}$ $(2 \times 10^{10} M_{\text{Pl}})$	$7.2 \times 10^{15}\,\text{GeV}$	only $N_c = 3$
$\beta = +1/\sqrt{3}$	$2.0 \times 10^{19}\,\text{GeV}$ $(1.64\, M_{\text{Pl}})$	$3.8 \times 10^{12}\,\text{GeV}$	only $N_c = 3$
$\beta = +1/\sqrt{3}$, $M_{Z'} \geq 4.6\,\text{TeV}$ (LHC)	$2.17\, M_{\text{Pl}}$	—	only $N_c = 3$

Table 5: Selection with one-loop threshold matching over the scanned spectrum. The last row restricts the scan to the dilepton bound of Appendix A.

The selection holds for any cutoff Λ between the best $N_c = 5$ pole and the worst $N_c = 3$ pole. Over all criteria ($g = 4\pi$ or $\alpha = 1$, with or without Yukawas), any Λ from 7.2×10^{15} to $1.4 \times 10^{19}\,\text{GeV}$ works. This window contains M_{Pl}, the reduced Planck mass and the conventional unification scale.

6 The scalar sector to the Planck scale

The selection requires every coupling of 3-3-1 at $N_c = 3$ to stay perturbative to M_{Pl}, so the scalar sector must pass the same test. We implement the general one-loop quartic and Yukawa RGEs [9, 10, 11] on the 18 real scalars; the engine reproduces the Standard Model β_λ and β_{y_t} exactly. The nine quartics allowed by the model's discrete symmetry close under the RGEs (residual 10^{-15}), and the potential is gauge invariant to 10^{-16}. The light-Higgs quartic is matched at 4 TeV to $\lambda_{\text{SM}} = 0.076$ at tree level in the decoupling limit. Survival means every $|\lambda| < 4\pi$, every Yukawa $< \sqrt{4\pi}$, and a potential bounded below, checked numerically at eight scales up to M_{Pl}.

3-3-1 at $N_c = 3$ survives in both variants, in a corner of parameter space: 17 of 250 viable points ($\beta = -1/\sqrt{3}$) and 13 of 250 ($\beta = +1/\sqrt{3}$) under small priors, none with $O(1)$ quartics. In the survivors the extra scalars cure the Standard Model's metastability. Consistency to M_{Pl} then requires the exotic Yukawa at 4 TeV to be below 0.80 ($\beta = -1/\sqrt{3}$) or 0.62 ($\beta = +1/\sqrt{3}$), i.e. exotic-quark masses below $1.79\, M_V$ or $1.40\, M_V$; the top's doublet must carry $c^2 \gtrsim 0.68$ of the electroweak vev squared; and all quartics at 4 TeV must be $\lesssim 0.37$. These statements are one-loop, with tree-level matching, and the search is a random sampling rather than a proof that the surviving region is unique.

7 Conditions and caveats

1. *The framework.* Everything assumes 3-3-1 with $\beta = \pm 1/\sqrt{3}$.

2. *Perturbativity to the Planck scale* is a standard criterion, not a theorem. A pole below $M_{\rm Pl}$ signals new physics rather than an inconsistency.

3. *Measured couplings are input.* Every N_c is run from $\sin^2\theta_W = 0.2504$, $1/\alpha_{\rm em} = 125.7$ and α_s at 4 TeV; in 3-3-1, $1/\alpha_X = (1 - 4s^2/3)/\alpha_{\rm em}$. For $\beta = -1/\sqrt{3}$, $N_c = 3$ reaches $M_{\rm Pl}$ for $\sin^2\theta_W < 0.449$, while $N_c = 5$ does so only for $0.066 < \sin^2\theta_W < 0.074$, a factor 3.1–3.4 below the measured value. For $\beta = +1/\sqrt{3}$, $N_c = 5$ fails at every weak angle, and $N_c = 3$ survives for $\sin^2\theta_W < 0.266$ (tree-level matching; +0.010 to +0.035 in margin with thresholds). The precise statement is therefore: *given the measured electroweak couplings*, only $N_c = 3$ stays perturbative to $M_{\rm Pl}$.

4. *Minimality is not explained.* Steps 1–2 leave $\{3, 5, 7, \dots\}$; the running does the selecting.

5. *No numbers follow.* The structure adds about eleven scalar parameters beyond the Standard Model's two and blocks gauge unification ($SU(3)_C$ and $SU(3)_L$ never meet).

8 Relation to prior work

That N_g is a multiple of N_c in 3-3-1 is known [1, 2, 3, 4]. Frampton obtained $N_c = 3$ by inputting three light neutrinos [6]. Charge quantization for arbitrary N_c was discussed in [14], generation counting for arbitrary β in [15], and the N_c-dependence of the charges in [16]. The closest predecessor is Doff and Pisano [5], who keep n_c general and find $N_g = k\,n_c$ for any $k \geq 1$; they do not use asymptotic freedom to set $k = 1$, nor running to choose n_c. Landau poles in 3-3-1 at $N_c = 3$ are studied in [17, 18, 19].

The step taken here is to keep N_c general throughout (charges, anomalies, scalars, two-loop running, thresholds, scalar sector) without taking the number of generations or neutrinos from experiment. Searches of arXiv and INSPIRE, including every paper citing the founding 3-3-1 papers, Frampton [6] and Doff–Pisano [5], found no prior use of asymptotic freedom plus running to fix N_c. This is not proof of novelty, and comments from experts are welcome.

9 Experimental tests

The $\beta = \pm 1/\sqrt{3}$ models predict a Z', singly or doubly charged bilepton gauge bosons and exotic quarks, which the LHC and future colliders will find or exclude. Dilepton limits on the Z' depend on the variant and on its decays. Published recasts use partial data: for $\beta = -1/\sqrt{3}$, $M_{Z'} > 4$ TeV from 36.1 fb^{-1} at 13 TeV with Standard Model decays only [20], weakened to 2.64 TeV with decays into exotic fermions open [21]; for $\beta = +1/\sqrt{3}$, the most recent dedicated recast we found uses 8 TeV data and gives $M_{Z'} \gtrsim 2.9$ TeV [22]. Appendix A recasts the ATLAS search with the full 139 fb^{-1} [26]:

$$\beta = +1/\sqrt{3}:\ M_{Z'} > 5.0\text{–}5.2\ \text{TeV}\ (4.6\text{–}4.9\ \text{TeV}), \qquad \beta = -1/\sqrt{3}:\ M_{Z'} > 4.7\text{–}5.0\ \text{TeV}\ (4.0\text{–}4.4\ \text{TeV}),$$

for decays to Standard Model fermions only (in brackets: with every exotic fermion open and massless). Since $M_V = M_{Z'}/1.225$, the bilepton gauge bosons then lie above about 3.3–4.2 TeV, and the bounds of Section 6 allow exotic quarks up to 1.40–1.79 M_V, so they may or may not be light enough for the Z' to decay into.

Consistency to $M_{\rm Pl}$ adds the relations of Section 6: exotic quarks lighter than 1.79 M_V or 1.40 M_V, $M_{Z'} = 1.225\,M_V$, and the top's doublet carrying $\gtrsim 70\%$ of the electroweak vev squared. A discovery violating these would break the selection without excluding 3-3-1.

In the minimal $\beta = -\sqrt{3}$ model, $\sin^2\theta_W$ reaches $\frac{1}{4}$ at 3.68 TeV at one loop, where g_X has its Landau pole, so its Z' must lie below about 3.7 TeV. The LHC bound in this model is $M_{Z'} > 3.7$ TeV (13 TeV, 36.1 fb^{-1}, exotic decays included) [23], and atomic parity violation

in caesium requires $M_{Z'} \gtrsim 4.2$–$5.4\,\text{TeV}$ [24]. The minimal model is therefore excluded unless higher-order effects raise its pole by more than about 14%.

Finally, N_c can be checked directly at low energy. With the N_c-dependent charges of Section 2 and the $O(\alpha_s)$ correction, $R = \sigma(e^+e^- \to \text{hadrons})/\sigma(e^+e^- \to \mu^+\mu^-)$ below charm is $N_c(Q_u^2 + 2Q_d^2)(1 + 3C_F\alpha_s/4\pi)$, which is 2.173 at $N_c = 3$. KEDR's weighted average over 1.84–3.05 GeV is $R = 2.225 \pm 0.020 \pm 0.047$ [25], so $N_c = 3$ agrees to 1.0σ, while $N_c = 2$ and 5 miss by 15σ and 33σ. This check is independent of the running argument; it is of course not new, but it confirms that the N_c-dependent charges reproduce the measured value.

A Recast of the ATLAS 139 fb^{-1} dilepton search

ATLAS published 95% CL upper limits on the fiducial $\sigma \times B(X \to \ell\ell)$, for one lepton flavour, for relative widths $\Gamma/M = 0$–10%, together with its own fiducial predictions for the benchmark Z'_{SSM}, Z'_ψ and Z'_χ [26]. We digitised these curves from the vector figures of the paper (Fig. 3, combined $ee + \mu\mu$); the digitised limits reproduce the values quoted in the text (3.6 and 13.1 fb at 250 GeV, 0.014 and 0.018 fb at 6 TeV, for zero and 10% width) to within 1–4%.

Method. In the narrow-width approximation, neglecting interference with Drell–Yan and Z–Z' mixing as the ATLAS benchmarks do, $\sigma B \propto \sum_q L_{q\bar{q}}(M)\,(g_{L,q}^2 + g_{R,q}^2) \times B(\ell\ell)$. We take the ATLAS Z'_{SSM} prediction and multiply it by the ratio of this quantity between the 3-3-1 Z' and Z'_{SSM}. Higher-order QCD corrections and the fiducial acceptance, the same for vector resonances to a few per cent, cancel in the ratio. Parton luminosities at $\mu = M$ use the PDF4LHC21 and NNPDF3.1 NNLO central sets. The mass limit is where the prediction crosses the observed limit at the model's width, interpolated between the published widths.

Couplings. With $X = Y - \beta T_8$, the heavy neutral boson orthogonal to hypercharge couples to each chiral field as

$$g_{Z'}(f) = \frac{g\,(T_8 - \beta\, t_W^2\, Y)}{\sqrt{1 - \beta^2 t_W^2}}, \tag{4}$$

with $T_8 = \pm 1/(2\sqrt{3})$ for SM doublets in triplets and anti-triplets and $T_8 = 0$ for singlets. This agrees with a direct diagonalisation to 10^{-16}, reproduces the known $\beta = -1/\sqrt{3}$ electron vector coupling $g(1-4s_W^2)/(4c_W\sqrt{3-4s_W^2})$, and for $\beta = -\sqrt{3}$ diverges at $s_W^2 = \frac{1}{4}$, the minimal model's Landau pole. Leptons sit in triplets; two quark generations sit in anti-triplets and one in the triplet. We take either the third (assignment A, the usual choice) or the first generation (B) in the triplet. Couplings are evaluated with $s_W^2 = 0.2504$ (4 TeV) and, as a check, 0.2312.

Validation. The procedure reproduces the benchmark widths (2.99, 0.53 and 1.16% for SSM, ψ, χ, against ATLAS's 3, 0.5 and 1.2%). Applied to the Z'_{SSM} curve with the E_6 couplings, it reproduces ATLAS's own Z'_ψ and Z'_χ predictions to within 2–6% between 3 and 5 TeV with PDF4LHC21, and their mass limits: 4.53 and 4.81 TeV against the published 4.5 and 4.8 TeV. With NNPDF3.1 the χ limit comes out at 4.96 TeV; the difference traces to the large-x antiquarks, where the two sets differ, so we use PDF4LHC21 as the baseline and quote the NNPDF3.1 spread. As an independent check, opening the exotic decays for $\beta = -1/\sqrt{3}$ reduces $B(\ell\ell)$ to 37% of its SM-only value, consistent with the reduction to 34% found in Ref. [21].

Results. Table 6 lists the observed limits. Over the quark assignment, the s_W^2 scheme and the PDF set, the limits move by about 0.2 TeV; opening every exotic decay lowers them by 0.3–0.7 TeV. Not included are Z' decays to bilepton or scalar pairs, which would add width and lower the limits further, and differences in higher-order corrections between models.

β	assignment	decays	Γ/M	$B(ee)$	observed	expected
$+1/\sqrt{3}$	A	SM	1.68%	6.58%	5.21 TeV	5.18 TeV
	A	SM + exotic	3.27%	3.38%	4.89 TeV	4.85 TeV
	B	SM	1.68%	6.58%	5.03 TeV	5.00 TeV
	B	SM + exotic	3.27%	3.38%	4.70 TeV	4.67 TeV
$-1/\sqrt{3}$	A	SM	1.15%	3.85%	4.79 TeV	4.75 TeV
	A	SM + exotic	3.09%	1.43%	4.04 TeV	4.23 TeV
	B	SM	1.15%	3.85%	4.96 TeV	4.93 TeV
	B	SM + exotic	3.09%	1.43%	4.39 TeV	4.42 TeV

Table 6: 95% CL lower limits on $M_{Z'}$ from the ATLAS 139 fb^{-1} dilepton search (PDF4LHC21, $s_W^2 = 0.2504$). A: third quark generation in the triplet; B: first. "SM + exotic": decays to the exotic quarks and heavy leptons open and kinematically unsuppressed.

Reproducibility

Each number above is produced by a script in `src/m331_*.py`, with reference output in `results/m331/` and regression tests in `tests/test_331_*.py`, run on every commit. The recast is `src/m331_zprime_recast.py`; the digitised ATLAS curves and the parton luminosities it reads are produced by the scripts in `tools/`.

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